Polar Equations and Graphs

ANSWER KEY

Recognizing polar graphs

1 Identify the graph of each polar equation:

a $r = 3$ Circle of radius 3 centered at the pole.

b $\theta = \frac{\pi}{6}$ Line through the pole at angle $\frac{\pi}{6}$.

c $r = 2 + 2\cos\theta$ Cardioid (heart-shaped, symmetric about the polar axis).

d $r = 4\sin 3\theta$ Rose curve with 3 petals.

2 For the limaçon $r = 1 + 2\cos\theta$:

a Explain how you know it has an inner loop. In $r = a + b\cos\theta$, $|a| < |b|$ (here $1 < 2$) produces an inner loop.

b Find the values of θ in $[0, 2\pi)$ at which the curve passes through the pole. $1 + 2\cos\theta = 0$ gives $\cos\theta = -\frac{1}{2}$: $\theta = \frac{2\pi}{3}, \frac{4\pi}{3}$.

3 For the rose $r = 5\sin 2\theta$:

a State the number of petals. $n = 2$ is even, so $2n = 4$ petals.

b State the maximum value of $|r|$ and the smallest positive θ at which it occurs. Maximum $|r| = 5$, first at $2\theta = \frac{\pi}{2}$, i.e. $\theta = \frac{\pi}{4}$.

4 Determine whether the graph of $r = 2 + \sin\theta$ passes through the pole, and state the type of limaçon.
$2 + \sin\theta \geq 1 > 0$ for all θ, so it never passes through the pole. Since $|a| > |b|$ $(2 > 1)$, it is a dimpled limaçon with no inner loop.

Symmetry tests

5 Test $r = 3\cos 2\theta$ for symmetry about the polar axis (replace θ with $-\theta$) and about the pole (replace r with $-r$).
Polar-axis: $\cos(2(-\theta)) = \cos(2\theta)$, unchanged — symmetric about the polar axis. Pole: replacing $r \to -r$ gives $-r = 3\cos 2\theta$, not equivalent to the original in general — the standard test is inconclusive for pole symmetry, though the curve (a 4-petal rose) is in fact symmetric about the pole by direct graphing.

6 Test $r = 2 + 2\sin\theta$ for symmetry about the line $\theta = \frac{\pi}{2}$ (replace θ with $\pi - \theta$).
$\sin(\pi - \theta) = \sin\theta$, so the equation is unchanged: symmetric about $\theta = \frac{\pi}{2}$ (the vertical axis).

Graphing from a table of values

7 For the cardioid $r = 1 + \cos\theta$, complete the table of r-values: $\theta = 0, \frac{\pi}{2}, \pi, \frac{3\pi}{2}$.

a $r(0) \ = 2$ **b** $r\left(\frac{\pi}{2}\right) \ = 1$ **c** $r(\pi) \ = 0$ **d** $r\left(\frac{3\pi}{2}\right) \ = 1$

8 For the circle $r = 4\sin\theta$, find the value(s) of θ in $[0, \pi)$ where r reaches its maximum, and state that maximum value.

Maximum of $\sin\theta$ is 1 at $\theta = \frac{\pi}{2}$, giving $r = 4$.

Rose curves in depth

9 For the rose $r = 3\cos 5\theta$, state the number of petals and the length of each petal (maximum $|r|$).

$n = 5$ is odd, so there are $n = 5$ petals, each of length $|r|_{\text{max}} = 3$.

10 Explain, using the general rule for rose curves $r = a\cos(n\theta)$ or $r = a\sin(n\theta)$, why n odd gives n petals but n even gives $2n$ petals.

When n is odd, tracing θ from 0 to 2π retraces each petal a second time (due to negative r values overlapping earlier petals), so only n distinct petals appear. When n is even, no such overlap occurs, and all $2n$ petals from the full $[0, 2\pi)$ sweep are distinct.